

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)**(TRIMESTER –III) END TERM EXAMINATION, APRIL -2024****Supply Chain Network Design****[Time: 3 Hours]****[Maximum Marks: 60]****Following are the course outcomes of the subject: - Supply Chain Network Design**

CO	Course Outcome (CO)
CO1	Understand the foundational concepts and principles of supply chain networks, emphasizing their significance in modern business operations.
CO2	Analyze the diverse factors influencing network design decisions, such as cost, service levels, and agility, to develop a holistic perspective on network management.
CO3	Identify the key factors to be considered when designing a distribution network

*(Note- Faculty can add the more CO's as per their subject.)***SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)**

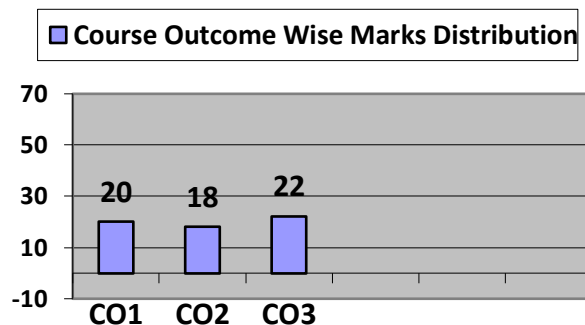
Q.1	Marks	CO
(A)	4	CO – 1
(B)	4	CO – 1
(C)	4	CO – 1
(D)	4	CO – 1
(E)	4	CO – 1

SECTION-B**Attempt all questions out of the following: (Long Answer Type Questions)**

Q. No	Marks	CO
2 (A)	6	CO – 2
2 (A)	6	CO – 2
2 (B)	6	CO – 2
2 (B)	6	CO – 2
3 (A)	6	CO – 2
3 (A)	6	CO – 2
3 (B)	6	CO – 3
3 (B)	6	CO – 3

SECTION-C**Attempt all parts of the following: (Caselet / Numerical)**

Q.4	Marks	CO
4 (A)	8	CO – 3
4 (B)	8	CO - 3



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Supply Chain Network Design

[Time: 3 Hours]

[Maximum Marks: 60]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A

(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following:

(5×4=20)

- (a) Supply chain network design is important for any business. Do you agree? Support your answer.
- (b) If a business wants to achieve more responsiveness, which type of network is suitable? Explain with reasons.
- (c) If a business wants to have high control of operation and integrated inventory, which type of network is suitable? Explain with reasons.
- (d) How does supply and allocation demand helps in supply chain network design?
- (e) Examine the role of technology in developing green supply chains.

SECTION-B

(Long Answer Type Questions)

Q.2. Attempt all questions of the following

(4×6=24)

- (a) You are designing a supply chain network, which factors would you consider that might affect the business network?

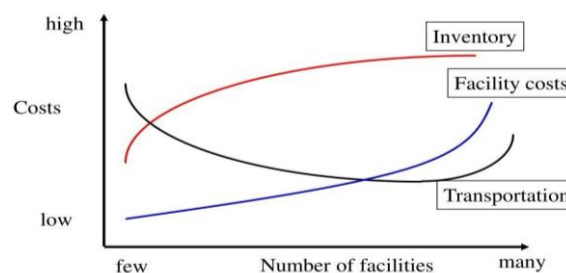
OR

- (a) You are working as supply chain manager for a start-up. Suggest the founder which type of network is suitable for the business. Explain the different characteristics of the chosen network.
- (b) While designing the supply chain network it is important to understand the importance of customer service components such as order visibility and return ability. How does these components affect the network cost and service level and agility? Explain.

OR

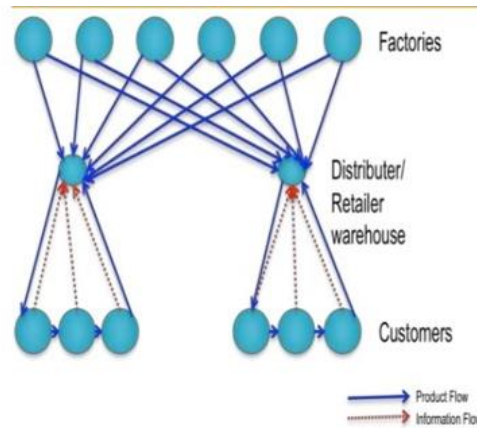
- (b) Your company is looking to outsource its manufacturing operation. The main goal is to achieve low cost and take advantage of better technology. As supply chain manager suggest and explain which outsource option is suitable for the business and why?

Costs and Number of Facilities



Q.3. (a) Explain the concept shared in the image.

OR



(a) Identify the distribution network and explain its cost and service factor.

(b) Explain any 3 Risk in global supply chain. Also mention the related mitigation strategy associated with them.

OR

(b) Natural disaster, Information infrastructure breakdown and Inflexibility of supply source are associated with which category of risk? Identify and explain.

SECTION-C

Turkish cotton is well known for its especially long fibers. It produces strong and soft threads when woven, and the textile produces superior absorbency. Towels made from Turkish cotton are extremely soft, light, and absorbent. They are much thinner than traditional towels, and they are therefore very light and packable for travel. Yara Textile is a name synonymous with high-quality Turkish towels in Turkey. The Yara family has owned and operated the business for two hundred years in Istanbul. The family has enjoyed an excellent reputation and healthy sales throughout the Middle East and Southeast Asia. The current president and owner, Mustafa Yara, has the vision of taking the Turkish towels to the North American market. However, it has proven challenging because of how competitive the market is. Yara's primary problem is distribution. How can Yara bring these luxurious Turkish towels from Istanbul to customers throughout North America? Mr. Mustafa Yara has assembled a supply chain team to tackle the distribution problem via three channels direct sale through online channels, wholesale through distributors, and wholesale through a consignment warehouse.

Direct sale to end consumer provides Yara the most control. Yara uses eBay and Etsy for direct consumer sales. When an order comes in, the Yara central warehouse packs and ships individual parcels through Turkish Post, crossing docking at US Postal Service and Canada Post to the end consumer. While the cost of shipping is relatively low, the lead time can be as long as 21 days for customers, causing some friction of purchase. The biggest advantage of this channel is the margin. Because there are no middle layers adding markups, the direct sales to consumers bring in an 80 percent margin for Yara. The volume is, however, minimal, hovering at around 20,000 units per year.

Traditional Sales through Distributors

This is the most traditional route to reach the masses but takes the most sales power. Because the textile space is competitive in North America, the Yara sales team engages local dealers to pitch to large textile distributors across North America. Yara pays the dealers a commission for each successful sale. The margin for this channel is as low as 5 percent. However, the volume of two million units per year makes up for the low margin.

There is another non-conventional wholesale channel for Yara. There are six third-party logistics (3PL) distribution centers in North America that carry their own home furnishing catalogs. The 3PLs partner with various retailers, like Pottery Barn and Home Sense, to distribute towels and bedding products to end customers. However, these 3PLs do not purchase any inventory because they operate consignment warehouses. Yara selected these 3PLs for the only purpose of accessing their catalogs. The exposure to various retailers was attractive enough for Yara to bare all inventory risks. Yara is the sole owner of their products in these warehouses. The company pays a monthly fee for storage plus a handling fee per order dispatched and receives payment from the 3PL only for actual orders sold. The sales volume ranges between 500,000 to one million units per year across all six 3PLs, and margin vary between 15 and 40 percent. This is because the margins are directly tied to the carrying costs. The longer Yara needs to hold the inventory before it is sold, the lower the margin it earns.

All three distribution channels have different advantages and disadvantages. Mustafa is keen to continue operating in three channels to diversify Yara's distribution risks.

Q.4. Attempt all parts of the following:

(2×8=16)

4(a) Identify and explain the different distribution network adopted by Yara Textile. Would you continue with the distribution strategy adopted by Yara textile? Give valid reasons for your answer

4(b) What strategies can Yara Textile implement to reduce the lead time in their direct sales channel through online platforms such as eBay and Etsy, given the current challenge of a 21-day delivery time potentially causing purchase friction among North American customers?